1. Setup

UniProt

- **20,431 Human reviewed proteins**
  - **Fasta files**
  - **Topology (annotation tables) → transmem, topo_dom, lipid, signal,** intramem**,** subcellular_location**,** carbohyd**, protein_existence.**
  - **xrefs** → just for cross reference of IDs.

→ 5,230 cross the membrane, but not fully annotated. **Need to run DeepTHMM**

DeepTMHMM

Run under academic license
Min 20 residues crossing membrane (dyn ths)

4,811	UniProt and DeepTHMM say membrane
419	UniProt only
150	DeepTMHMM only
15,051	neither

\+ GPI anchors from UniProt annotation
\- Internal membrane proteins

S1	3,385	both sources agree, ectodomain >19 aa
S2	110	one source only, ectodomain >19 aa
S3	130	GPI-anchored
Flagged	516	membrane evidence, ectodomain <20

\+ CSPA annotation
\+ SURFY annotation

Human Protein Atlas (HPA)

- **20,162 genes**
  - **proteinatlas.tsv → for symbol lookup**
  - **rna_tissue_hpa.tsv** → 20,162 genes x 40 tissues. Independent reading of healthy tissue (**nTPM**).
  - **subcellular_location.tsv** → 13,603 genes (IFM). Plasma membrane 1,648. Exclude only on positive evidence of being internal-only. Used to “rescue” internal only / unannotated UniProt.

UCSC Xena Toil

- **TCGA ~** 11,000 tumors across 33 cancer types + adjacent normal tissue
- **GTEx ~** 50 body sites, healthy tissue from deceased donors

Toil reprocessed raw reads with same pipeline

- **data matrix** + phenotype metadata + IDs mapping:
  ~60K genes, ~19K samples

- **9,185 primary tumors**
- **32 cancer types**
- **727 TCGA adjacent normals**
- **7,429 GTEx healthy samples across 51 tissues**

Clinical Proteomic Tumor Analysis Consortium (CPTAC)

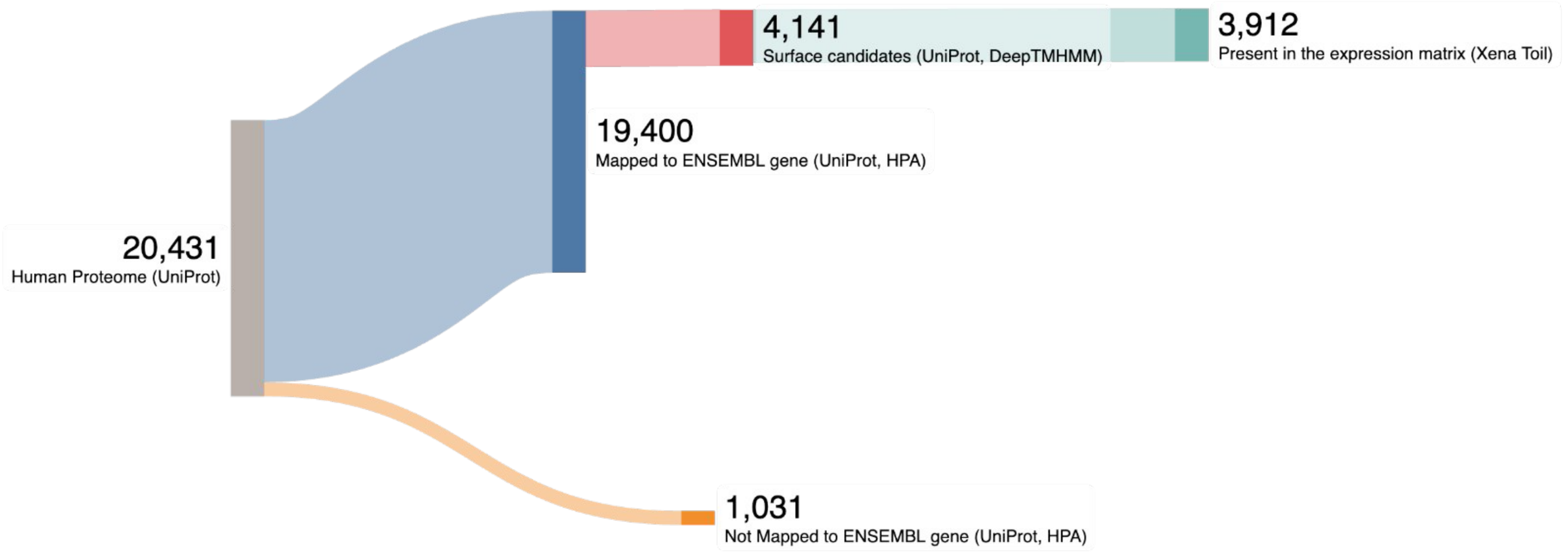
- **Mass spec from same cancer types TCGA RNA-sequenced**
  - 28 files → **Only 8 cohorts have tumor and matched normal measurements.**
  - **Mass spec under detects hydrophobic membrane proteins → used for reporting not for ranking.**

Hugo Gene Nomenclature Committee (HGNC)

- **Used for gene groups → to inform report.**

2. Pipeline

Which genes are we going to study?



Which of those are “up” in tumor vs healthy?

1) Map each tumor type to its healthy tissue counterpart.

- **Match TCGA 32 cancer types to GTEx healthy tissues.** First, Wang et al. 2018 (19 cancer types)
- **Organ in GTEx? Keep**
- **Right organ wrong compartment? keep, grade imperfect**
- **No organ? Find tissue that contains same cell types, grade lineage only**
- **Nothing contains that cell type? Drop it** (Thymoma)

2) Run the comparison

- **Tumor vs Healthy (GTEx) + Tumor vs Adjacent normal (TCGA)**
- If any of the **groups <10 samples, don’t run comparison. Total possible comparisons: 46 (30 cancer types, 15 have both, 15 only GTEx + bladder)**
- **Wilcoxon test:** one per gene, per comparison = 46 × 3,912.
- **Per comparison (within cancer type), correct FDR**
- **FDR < 0.01 and log2FC ≥ 2**

- **1,513 genes pass**

Across the whole body, what is the highest this gene goes in any healthy tissue?

1) Define S score as:

- **S = log2(tumor median TPM + 1) - log2(highest healthy tissue TPM +1)**
- **Calculate it for every gene, for every cancer type.** 4 different variants.
- Take the **MAX(S)** as score for **pan-cancer ranking**.

2) Ranking:

- **Highest S to lowest per cancer type.**
- **Global top 10 (top, max 2 per family)**
- **Pan cancer top 10 (which gene is raised in the most cancer types).** Count how many distinct cancer type the gene passes in. Compute median, min, max S across types. Sort by number of cancer types, S as tiebreak

How robust are the results?

1) Re running the pipeline under 5 different threshold changes:

- **Minimum extracellular segment**
- **FDR**
- **log2 fold change**
- **healthy reference**
- **healthy summary**

3. Results